>_ CYBER TECH ACADEMY

Cybersecurity Essentials

Core security principles, threats, risk, identity, cryptography, scams, and safe digital habits.

LEVEL: BEGINNER

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How to use this guide

Read the concepts first, run examples only in a system you own or are authorised to use, complete the lab, then answer the review questions without looking back. Commands and outputs can differ by Linux distribution, software version, and local configuration.

Responsible learning: only test systems for which you have explicit permission. These notes are educational and do not grant permission to access, scan, disrupt, or modify third-party systems.

Learning objectives

- Explain confidentiality, integrity, and availability.
- Distinguish threats, vulnerabilities, likelihood, impact, and risk.
- Recognise common social-engineering patterns.
- Explain authentication, authorisation, MFA, encryption, and hashing.
- Recommend practical controls for personal and organisational security.

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1. What cybersecurity protects

Cybersecurity protects systems, networks, applications, and information against unauthorised access, misuse, disruption, modification, or destruction. It includes people, processes, and technology. A strong technical control can still fail if people do not understand how to use it or if procedures are unclear.

The CIA triad is a foundational model: confidentiality means only authorised parties can access information; integrity means information remains accurate and changes are authorised; availability means systems and data are accessible when needed.

Principle	Example failure	Possible control
Confidentiality	Private file exposed publicly	Access control, encryption, data minimisation
Integrity	Unauthorised change to a record	Hashes, signatures, permissions, audit logs
Availability	Service outage or ransomware	Backups, redundancy, recovery testing

Security is not only about attackers. Misconfiguration, mistakes, equipment failure, natural disasters, and third-party outages can also create incidents.

2. Threats, vulnerabilities, and risk

A threat is a potential cause of harm. A vulnerability is a weakness that could be exploited or triggered. Risk combines the likelihood of an event with its potential impact. A control reduces likelihood, impact, or both. Risk ratings are estimates to support decisions, not exact scientific measurements.

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Risk discussion checklist:
1. What asset needs protection?
2. What could happen to it?
3. Which weakness or condition makes it possible?
4. How likely is the event?
5. What would the impact be?
6. Which control reduces the risk?
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Use a simple risk register to document the asset, scenario, existing controls, likelihood, impact, owner, treatment, and review date. Common treatments include mitigating the risk, avoiding the activity, transferring some financial impact, or accepting the risk with appropriate approval.

3. Identity, passwords, and account security

Identification is claiming an identity; authentication verifies it; authorisation decides what that identity may do. Multi-factor authentication combines different factor types, such as something you know, something you have, and something you are. Two passwords are not two different factor types.

Use unique passwords for each service, preferably generated and stored by a reputable password manager. Enable MFA, prioritising email, financial, cloud, and administrator accounts. Keep recovery codes securely, review active sessions, and remove access that is no longer needed.

Credential stuffing uses previously exposed username/password pairs against other services. Password reuse makes it effective. Rate limiting, MFA, breached-password screening, secure password hashing, and suspicious-login monitoring help reduce risk.

Legitimate security teams do not need your password or MFA code. Never share one-time codes with callers or people in chats, even if they claim to be support staff.

4. Encryption, hashing, and signatures

Encryption transforms readable plaintext into ciphertext using a key so that authorised parties can recover the data. Symmetric encryption uses the same secret key for encryption and decryption. Asymmetric cryptography uses a public/private key pair and supports uses such as key exchange and digital signatures.

Hashing maps data to a fixed-length digest. A cryptographic hash is designed to make it impractical to recover the original input or find a different input with the same digest. Hashes are useful for integrity checks, but an ordinary hash is not a password-storage solution by itself.

Password storage should use a purpose-built password-hashing function such as Argon2id, scrypt, or bcrypt with a unique salt and suitable cost parameters. Digital signatures help verify origin and integrity; they do not automatically make the signed content safe.

Mechanism	Main purpose	Important distinction
Encryption	Confidentiality	Reversible with the right key.
Hashing	Integrity/fingerprints	Designed to be one-way.
Digital signature	Origin and integrity	Uses private-key signing and public-key verification.
Encoding	Data representation	Not a security control by itself.

5. Social engineering and online scams

Social engineering manipulates people into revealing information, sending money, or performing unsafe actions. Common tactics include urgency, authority, fear, curiosity, fake delivery notices, investment promises, job scams, impersonation, and requests to bypass normal procedures.

Check the full sender address and domain, not just the display name. Navigate to important services using a known bookmark or typed address instead of unexpected links. Verify payment or account-change requests using a second trusted channel. Do not open unexpected attachments on a system containing sensitive data.

HTTPS and a padlock indicate an encrypted connection to a domain, not that the domain is trustworthy. Scammers can use HTTPS too. If an account may be compromised, change its password from a trusted device, revoke active sessions, enable MFA, and contact the service using official channels.

6. Backups, updates, and security habits

Install operating-system and application security updates from trusted sources. Remove software you no longer need, use standard accounts for everyday work, lock devices when unattended, and encrypt sensitive devices where supported.

Backups should be protected from the same event that could destroy the original data. Keep separate or offline copies where appropriate and test restoration. A backup that has never been restored is an assumption, not a proven recovery plan.

Report suspicious events promptly. Preserve relevant messages and timestamps, avoid deleting evidence, and follow organisational procedures. A fast, accurate report often reduces the impact of an incident.

Guided practical lab

Exercise: assess five fictional scenarios and recommend a proportionate control.

- An employee reuses a password on work and personal accounts.
- A laptop containing customer records is lost.
- A payment request arrives with an urgent deadline and a changed bank account.
- A web service has no tested backups.
- A public file-sharing link exposes internal documents.

Write each scenario as: asset → threat/event → weakness → impact → control → verification method. For example, verify a backup control by performing a controlled restoration test, not merely by checking that a backup job says 'success'.

Mini exercise: create a personal account-security checklist that includes a password manager, MFA, recovery methods, software updates, and backup checks.

Knowledge check

Try to answer these before checking the answer key. Write your reasoning, not just a single word.

1. What are the three parts of the CIA triad?

Answer: _____

2. How is authentication different from authorisation?

Answer: _____

3. Why is a normal fast hash unsuitable by itself for storing passwords?

Answer: _____

4. Does HTTPS prove a website is legitimate?

Answer: _____

5. What should you do with a suspicious payment change request?

Answer: _____

Answer key and reasoning

1. Confidentiality, integrity, and availability.

2. Authentication verifies who or what an entity is; authorisation determines permitted actions.

3. Attackers can test guesses rapidly. Use a dedicated password-hashing function with a unique salt and appropriate cost.

4. No. It encrypts the connection to the domain, but a malicious site can also use HTTPS.

5. Verify it through a separate trusted channel and follow established approval procedures.

Glossary

Term	Meaning
CIA triad	Confidentiality, integrity, availability.
Threat	Potential cause of harm.
Vulnerability	Weakness that may be exploited or triggered.
Risk	Likelihood and impact considered together.
MFA	Authentication using multiple distinct factor types.
Salt	Unique random value added when hashing a password.
Social engineering	Manipulating people into unsafe actions or disclosures.

Further study

Continue with the next Cyber Tech Academy module, practise in a disposable virtual machine or an explicitly authorised lab, and consult official manuals for the tools you use. Verify commands against your own system before running them.

Cyber Tech Academy • Free learning • Learn the theory. Practise safely. Build real skills.